

# Exploring Affordable Mixed Reality Fire Pump Training Using a COTS Head-Mounted Display and Passive Haptics

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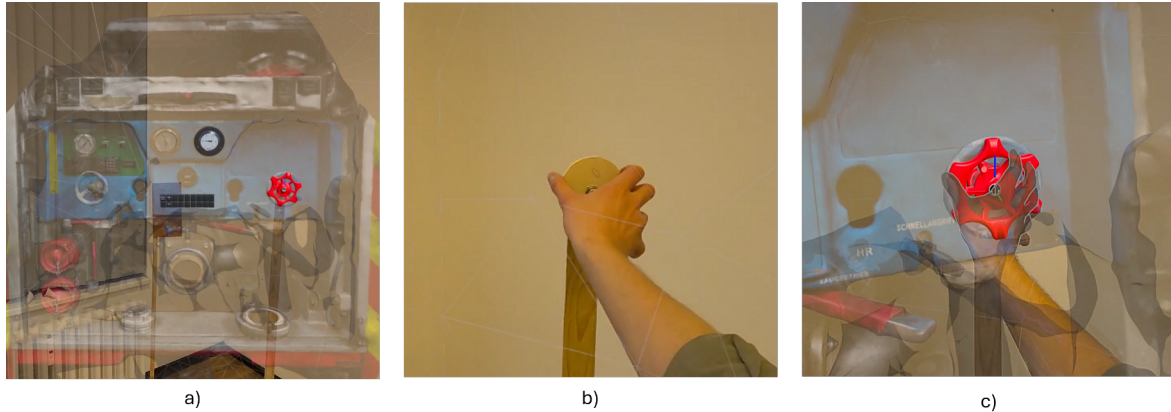


Figure 1: (a) A 3D reconstruction of a fire pump combined with interactable and animated 3D objects, is aligned with a low-cost physical prop in Mixed Reality using a Meta Quest 3. (b) The wooden prop provides haptic feedback during (c) hand-interaction with the virtual replica. Hand tracking of the Meta Quest 3 is used for interaction with virtual objects.

## ABSTRACT

Simulation-based training is increasingly utilized because training professionals on real equipment can be costly and time-consuming, requiring the removal of machines and environments from operational use. Advanced Virtual Reality (VR) and Mixed Reality (MR) simulators, though offering enhanced interaction, are often complex, costly, and difficult to maintain. Partnering with the Cologne fire department, we are developing an MR simulator for fire pump training using commercial off-the-shelf (COTS) solutions to tackle those challenges. We use a Meta Quest 3 head-mounted display to visualize a virtual replica of the fire pump, combined with wooden physical props to provide passive haptic feedback. This paper documents the development process and discusses initial findings, i.e., concerning spatial alignment without external visual tracking and regarding hand tracking based interaction. We propose a resulting research agenda for this approach and discuss opportunities and limitations to amplify user and training experience through affordable MR.

**Index Terms:** Haptic interfaces, human-computer interaction, mixed / augmented reality, virtual training.

## 1 INTRODUCTION

Training of professionals is often expensive and time intensive, as it can block equipment, machinery, and environments from regular use at work. Therefore, simulations of training are often used. Simulators have gained a new dimension of perception and interaction capabilities with the re-emerging XR field. Still, a lot of those

systems that combine Virtual Reality (VR) and other sensor-based simulator systems in a Mixed Reality (MR) setting are complex to set up, need a lot of maintenance and are expensive. We work with the fire department in Cologne, Germany, and try to develop an MR simulation of certain training steps, namely, a fire pump training. This type of fire pump is part of a regular fire truck, so training on the pump excludes the vehicle from firefighting operations. Simulating fire pump training separately would prevent this and mitigate material wear during training, so that the fire truck only has to be used in the final steps of training, while the previous steps can be carried out in the training application. Usage of the fire pump includes opening valves and testing water throughput, a task where manual interaction with physical objects and their reaction is relevant, making it difficult to simulate by just relying on non-haptic VR. Our approach was to develop a prototype of an MR system that would use low-cost and commercial off-the-shelf (COTS) solutions in order to make the system easy to setup, maintain and keep it low-cost. In the course of a university project with students, we investigated how COTS MR head-mounted displays (HMDs) combined with physical props for haptic feedback would perform in such a task. The main advantage of this approach over rebuilding a fully functional fire pump prop is that a virtual replica with low-cost haptics is cheaper, particularly when scaling up quantities. During the project, we used low-cost materials such as wood and screws etc. to build a physical prop for passive haptics (see Figs. 1 and 2). As HMD, we used a Meta Quest 3. We report on first concrete experiences with developing this setting and offer our learnings for discussion in the workshop.

## 2 RELATED WORK

Related work in the haptically enhanced MR provides a variety of systems that can be classified with the help of the taxonomy provided by Bouzbib et al. [1]: on the proposed matrix of *degree of physicality* and *degree of actuation*, our system has no actuation but high physicality. The prototype system in this paper relates to

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passive haptics MR systems as in the works of Hettiarachchi and Wigdor [4] and Zhou et al. [9]. The former paper investigates how everyday objects can be used and tracked, in order to add haptics to virtual objects in AR, based on factors like a similar geometry. Here, object tracking is used to align virtual and physical objects, not hand tracking. In the latter system, an everyday object is used as tangible user interface (TUIs) and the object is selected based on how the geometry or usage affordance fits to certain interactions like typing, swiping, etc. [9]. Here, hand tracking is implemented using gloves with optical marker-based motion capture. Both systems use freely movable everyday objects, not objects like a valve, that can only be rotated along an axis. Both works, as well as our paper, further relate to the concept of opportunistic controls [3] that proposes to repurpose the environment opportunistically as TUI. As an additional related focus, previous work [2, 8] explored how low-cost or even everyday objects can be repurposed to improve the experience in XR systems, based on their affordance related to the intended interaction. Our approach is to integrate low-fidelity and simple haptic props with an alignment solely relying on the hand tracking that comes with COTS HMDs to build a low-cost environment that fulfills the affordance needs for haptic feedback of a concrete virtual twin (i.e. the virtual fire pump).

### 3 METHODS

We explored the development of our system in an iterative student project together with two undergraduate students in an interdisciplinary curriculum involving computer science, design and business. We describe soft- and hardware, the virtual replica and the physical prop, as well as the spatial alignment and the interaction with it. Our MR system is developed using the Unity Game Engine (LTS 2022.3.33f1) along with the Meta XR SDK (65.0.0). See Fig. 3 for the architecture diagram. The application is deployed as a standalone app for the Meta Quest 3 HMD (MQ3). The systems works without an external computer and is thus aimed at being easier to setup and more mobile than stationary computer systems that involve external trackers.

We used Polycam 3D reconstruction software to create a virtual replica of the fire pump. The virtual replica is used as visual overlay in the HMD. We prototyped simple, affordable props made out of wood, screws and nuts (see Fig. 2). Since the 3D reconstruction of the fire pump is only a textured mesh, the interactive parts, like buttons, levers and valves, were modeled using Blender 3D software and incorporated into the fire pump model as animated and interactable parts.

To spatially align the virtual replica with the physical props, our system uses spatial anchors, as offered by the Meta XR SDK. Similar systems that use more sophisticated setups use optical tracking and/or sensors to map the virtual replica of the fire pump onto the position of the physical prop. However, the MQ3, like many other COTS HMDs, does not provide access to the camera stream of the headset and thus also does not enable tracking technologies such as image-, marker-, or object tracking. Our approach is instead to use virtual spatial anchors that can be placed manually in the real world environment using the MQ3 controllers (see Fig. 2). The controllers are only used in the first step of the application to place the spatial anchors. Afterwards, the controllers are left on a nearby table and the MQ3 automatically switches to hand tracking for MR interaction with the virtual buttons, valves, etc. Our system uses the built-in MQ3-native hand tracking to simulate realistic hand interactions and enable passive haptics with real-world objects.

### 4 CURRENT STATUS

In the current status of the project, two main interactions of the training are implemented as a prototype. First, the valve can be opened and closed through a rotating manipulation. The virtual valve model rotates along with the rotation of the tracked hands of

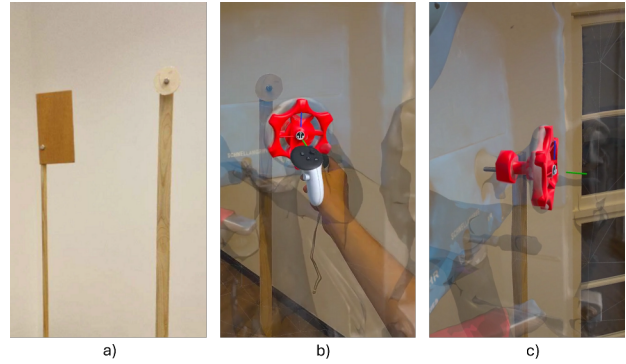


Figure 2: (a) The physical prop made out of wood, screws and nuts, with interactable parts for the button panel (left) and a valve (right); (b) The alignment process using the controller to set the spatial anchor at the valves position - completed in (c).

the user, which at the same time rotate the physical prop in the real world (see Fig. 1). The coupling of the real and virtual object is therefore implemented indirectly through hand tracking, as in the work of Zhou et al. [9].

The second interaction is pressing buttons to start the motor or increase or decrease the water pressure (see Fig. 4). The barometer is animated to respond to button presses, meaning that when the "Gas +" button is pressed, the pressure in the barometer increases, while the pressure decreases when the "Gas -" button is pressed. In training, the goal is to keep the pressure at 7 bar. The virtual buttons are visually overlaid on a wooden board. So when the user presses a button, they feel haptic feedback from the wooden board when the button is pressed through. The wooden board is attached to a flexible wooden stick, allowing some displacement in the direction of the button press.

The following subsections discuss the current status of interactions with the valve and buttons as well as with the overlay and inform about the challenges and our insights.

#### 4.1 Valve

The quality of the valve interaction is very dependent on the implementation of the grasping interaction with the hands. There are several ways in the Meta XR SDK to grab and manipulate objects. Our first iteration uses the so-called "TouchHandGrabInteractable". This component divides the object (the valve) into several colliders. When a finger of the user hits such a collider, the finger of the virtual hand is snapped into a certain position in that collider. This means that the grasping of an object does not have predefined hand poses but predefined attach points for fingers when the user grabs it. This helps to prevent the virtual hands from grabbing "through" an object. Moreover, the fingers attach to the surface of an object. It can also help in designing the natural grabbing of an object.

In the first iteration, the object has one collider that is not subdivided further. Therefore, the virtual fingers do not snap to predefined positions but rely on the objects surface where hit, while still being prevented from moving through the object. The main problem with this method is that it could not reliably determine when the user would release the valve. The object was grabbed as long as more than one finger was still in contact with the valve. However, a valve that allows for multiple full rotations has to be rotated by clutching out from a grab to reorientate the hand and then to re-engage by clutching back into the grab, in order to deal with the rotational limits of the arm and wrist joints. During such a clutching action, there is hardly any visible release of the hand from the valve. Sometimes, the hand even touches the valve and slides over

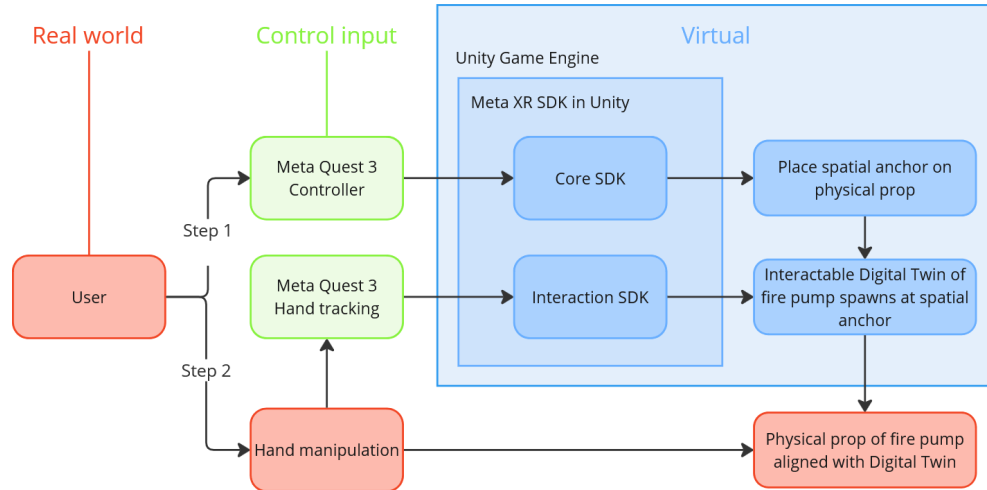


Figure 3: The architecture of our affordable MR approach describes how a user first (Step 1) places a spatial anchor to align a virtual replica with a physical prop, and secondly (Step 2), interacts with both virtual replica and physical prop using hand manipulation. The virtual system is based on the Unity game engine and the Meta XR SDK.

its surface without causing any movement of the valve. Even at the final release, the fingers of the hands only make micro-movements to release the valve. The only macro-movement is to disengage by retracting the whole hand from the valve.

As to the precision of the tracking, we experienced a high dependence on the SDK version as the hand tracking in the applied version of the Meta XR SDK was less accurate than the one in the newer Version 69.0.1. As a result, the detection of micromovements as a release action was very error prone and led to unsatisfactory interactions with the valve. For example, the valve would rotate back and forth, because the event in which the user's hand releases the valve was not detected correctly. An alternative approach could be to use pre-recorded hand poses that record a release when the hand pose is not performed anymore (with a certain error threshold). However, erroneous recognition of micromovements with relatively imprecise hand tracking still remains a problem.

Another problem arises from visual occlusion with hand tracking. The fingers can be occluded by the back of the hand while grasping, but also by the wooden structure around which the fingers grasp (see Fig. 1b). This issue was not experienced in our setting with the button interaction.

## 4.2 Buttons

The first iterations use the button implementations from the Meta XR SDK and we changed minor parameters and the visuals to match digital and physical buttons (see Fig. 4). When the virtual button is pressed using the index finger, the virtual button reacts like a real button: Upon contact with the virtual fingertip, the button moves alongside the real fingertip and stops to go any further when reaching the button base. Physically, the wooden board is aligned with the button surface, and the maximum depth displacement through bending roughly coincides with the button base. Overall, the button interactions work well as long as single button presses are induced. The current physical setup causes the wooden board to rebound in the direction of the finger after release from a press, possibly causing interference with quicker series of button presses.

## 4.3 Spatial alignment

As described in Section 3, the spatial anchor is used to place the virtual fire pump inside the real room and overlay the physical prop.

In our prototype, the spatial anchor is displayed as a 2D disk and local coordinate system and moves along the tip of the controller, with the surface forward vector pointing towards the center of the controller. The orientation of the virtual fire pump is aligned accordingly so that the front-side is oriented towards the center of the controller. The spatial anchor is located in the center of the valve (see Fig. 2). This helps the user match the position and orientation of the virtual and the physical object.

It also helps to anchor where haptic feedback should happen. During placement, the user can set the tip of the controller against the physical prop and tilt the anchor accordingly to align the orientation of the fire pump and the physical prop. When the placement is finished, the visualization of the spatial anchor should disappear and only the fire pump should remain. To make testing easier during development, the spatial anchor visualization was not removed after spawn in our prototype. By pressing a reset button on the controller, the fire pump model is deleted and the placement process starts again. A second method is to move the physical props physically to match the fire pump. This is however also a disadvantage of this solution, because unintended movement automatically leads to a mismatch in alignment between the firepump and the physical prop. The system is unable to automatically detect and handle such a mismatch without external tracking. The spatial anchor cannot be used for tracking, because it is static and would not move with the real-world object it was placed upon. Such a manual placing is also quite imprecise and depends on the user's competence with placing the spatial anchor. On the other hand, the system does only rely on a stand-alone COTS HMD and low-cost physical props without any additional sensory. The following section discusses how future research can contribute in identifying, evaluating determining possible sweet spots for this approach.

## 5 DISCUSSION OF FUTURE GOALS

### 5.1 Other possible approaches using the Quest 3

We faced several challenges during this first approach: The imprecision of hand tracking and interaction, imprecise placing of the virtual replica using spatial anchors the resulting erroneous alignment, also caused by the missing tracking of the physical objects. For a better initial alignment of the virtual replica and the physical prop using spatial anchors, several improvements can be made.



Figure 4: The user presses the virtual button using the index finger and pushes the physical prop upon contact leading to haptic feedback.

First, we can lock the horizontal rotation axis of the virtual replica to prevent the user from tilting it.

Second, the whole virtual replica, and especially the valve, needs to be more transparent. Currently, aligning it with the physical prop is difficult because the opaque valve occludes the physical prop, thus hindering the user from seeing the physical prop.

A further improvement, not only during the alignment phase but also for the interaction with the fire pump, would be to implement a depth-based cutout of the user's body, especially hands and arms. This could be implemented using the Depth API that has recently been added to the Meta XR SDK. The depth data could also be used to collect information about the position of the physical prop during the alignment phase, potentially eliminating the need for a controller.

Compared to tracking methods such as marker or sensor-based tracking, however, this method would still not be as precise. Other approaches for alignment would be to use the room mesh data or persistent semantic room data. However, both approaches rely on persistent data that are not regenerated at run-time. This means that when the physical props are moved, the data have to be manually regenerated outside of the app. The only other viable method for alignment is to use the depth information that is dynamically generated by the Depth API. Future work should also look into other COTS HMDs like Pico headsets that provide ArUco marker tracking implementations in their SDK.

## 5.2 Research agenda

In addition to the aforementioned aspects of improving visualisation and alignment, we seek to better understand the sweet spot of providing an easy-to-use COTS-based system for haptically enhanced MR training with a simple technical setup, while providing sufficiently accurate and believable interaction to allow users to effectively transfer acquired experience to the real fire pump system.

1. **Improved hand input precision:** We want to explore how transparent props, e.g., made of plexiglass, can reduce visual occlusion for the HMD-based hand tracking. In addition, we plan to improve flexibility with subdividing the collider object or by using predefined gestures.
2. **Additional tracking from outside:** In addition to the implementation and exploration of the Depth API, we want to explore low-cost tracking systems/sensors like potentiometers to sense the rotation of the valve and thus improve its tracking.

We want to incrementally add other low-cost hardware. In this regard we explore how it performs against other systems in the domain with respect to usability, complexity, cost and simplicity to install it.

3. **Improved training experience through visual delegates:** Misztal et al. investigated the impact of visual delegates on the user experience in games and VR applications [6, 7]. We want to study how those insights translate into the MR context and how they can improve the experience for trainees in such systems as ours. More concretely, during working on the pump, firefighters have to constantly re-assess the waterflow through the pipes, i.e., by laying their hands on the pipe and feeling vibrations. Instead of implementing haptic feedback through vibration motors, we aim at using visual delegation to create visual effects that imply the according proprioception and help training such manual steps during the handling of the pump.
4. **Complex training system with deeper functionality:** We plan to deepen cooperation with the fire department to create didactic settings and scenarios for a meaningful training on the fire pump and to prioritize which technical features should be implemented in what detail until a user evaluation of the training experience is possible.
5. **User Experience evaluation:** As next steps, we plan to conduct user studies to assess expert views on the training with the prototype. As reported in similar virtual training studies, a high training experience often correlates with a high presence and a high usability experienced by users [5], which should be measured accordingly, e.g., by using self-reporting tools such as the System Usability Scale, MEC-SPQ presence questionnaires, or the Training Evaluation Inventory. In addition, usability and cognitive load with setting up the system should be evaluated, e.g., by using NASA-TLX.
6. **Scalability of physical props:** Our low-cost approach aims at keeping hard- and software costs at a minimum, so we plan to explore how our wooden structures can easily be reproduced in scale by using other materials and prototyping methods, such as 3D printing or laser cutting.

## 6 CONCLUSIONS

This paper proposes the use of low-cost physical props in combination with COTS HMDs for haptically enhanced training of a fire pump. The fire pump serves as a concrete example for training on machinery where haptic manipulation is crucial for effective utilization. We reported on our current technical status as explored in a University student project and share our insights on the practical implementation using a Meta Quest 3 HMD and a Unity game engine-based software, solely relying on hand tracked input for object tracking. We discuss these findings along with a research agenda on improving hand tracking precision, additional sensor-support, using visual delegation, the didactic perspective, system scalability, and a proposed evaluation methodology, aiming to find a sweet spot between user precision, installation effort, affordability, and maintenance complexity.

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